RIFT tutorial

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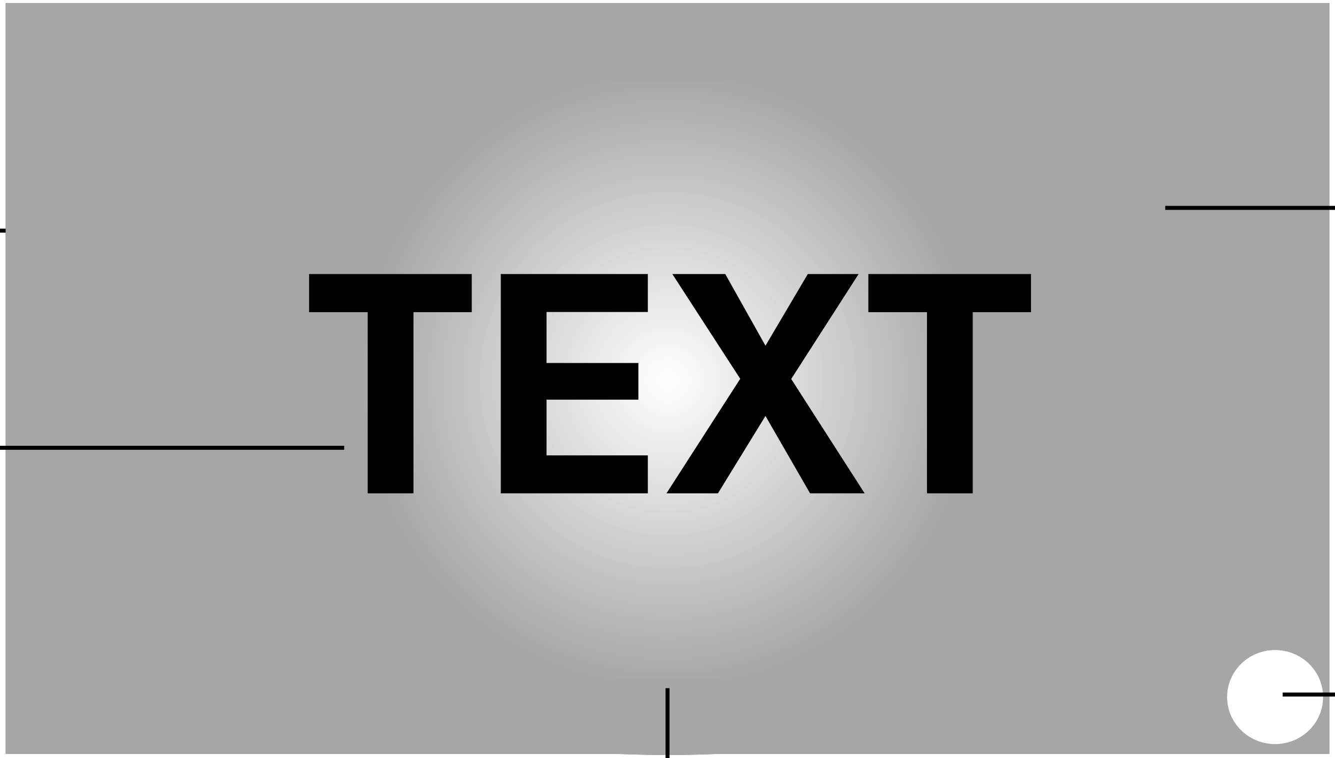
How to frequency tag a (visual) stimulus?

1. Draw a mid-grey background

4. Draw the visual stimulus



3. Apply a Gaussian mask to smooth sharp luminance transitions around the edges, ensuring the tagging area remains invisible across saccades

5. Draw a disk for recording the tagging signal with a photodiode

2. Flicker the area beneath the stimulus: the luminance of all pixels in this area follows the tagging signal sequence

The order of layers matters!

What are the steps of data analysis?

S1_Get_ExpInfo.m — Bad sensors, triggers, experimental design

S2_PreProcessing.m

- **Get_MEGData.m** — Data, header, MEG triggers
- **ft_preprocessing.m** — Remove bad sensors, bandpass filter ([0.5 100]Hz) & notch filter ([49 51]Hz)
- **ICA4rawdata.m** — Artefact rejection by ICA (oculomotor & heartbeat)
- **Get_EyeData.m** — Extract EM events and triggers from eye link file, "locate" EM events with sentence ID & word ID
- **Get_Event.m** — Align MEG triggers and EM triggers, create events using experiment design and EM events (wrt MEG)

Event.event_raw_header

	1	2	3	4	5	6	7	8	9	10
1	sentence_id	word_loc	loc2targ	SentenceCondition	fixation_on_MEG	saccade2this_duration	fixation_duration	NextOrder	FirstPassFix	PreviousOrder

Experiment design (columns 1–4)

Triggers to epoch MEG data (column 5)

Eye movement measures to select events (columns 6–10)

What are the steps of data analysis?

Script	Function	Description
S1_Get_ExpInfo.m		Bad sensors, triggers, experimental design
S2_PreProcessing.m	Get_MEGData.m	Data, header, MEG triggers
	ft_preprocessing.m	Remove bad sensors, bandpass filter ([0.5 100]Hz) & notch filter ([49 51]Hz)
	ICA4rawdata.m	Artefact rejection by ICA (oculomotor & heartbeat)
	Get_EyeData.m	Extract EM events and triggers from eye link file, “locate” EM events with sentence ID & word ID
	Get_Event.m	Align MEG triggers and EM triggers, create events using experiment design and EM events (wrt MEG)
S3_Get_all_epoches.m	Get_Epoch.m	Epoch data and remove bad trials
Coh_RIFT.m		Quantify tagging responses by calculating coherence between the MEG sensor & the photodiode (normalized amplitude over each trial)
Group_Coh.m		Perform group-level statistics by comparing conditional differences using a permutation test

Coh_RIFT.m

How to measure time-resolved tagging response?

1. Select the tagging response sensors

- Tagging response: coherence between a MEG sensor and the photodiode at the tagging frequency band
- Loop over all sensors (in the occipital cortex), over all trials (regardless of conditions)
- Compare the coherence during the time-of-interest (e.g., the pre-target fixation interval) and the baseline (i.e., no flicker interval)
- ft_statfun_indepsamplesZcoh.m

(one tail, alpha=0.01, 2000 permutations)

$$Z = \frac{(\tanh^{-1}(|\mathrm{coh}_1|) - \mathrm{bias}_1) - (\tanh^{-1}(|\mathrm{coh}_2|) - \mathrm{bias}_2)}{\sqrt{\mathrm{bias}_1 + \mathrm{bias}_2}}$$

$$\mathrm{bias}_1 = \frac{1}{2n_1 - 2}, \mathrm{bias}_2 = \frac{1}{2n_2 - 2}$$

where coh_1 and coh_2 denote the coherence value for the pre-target and baseline segments, bias_1, and bias_2 is the term used to correct for the bias from trial numbers of the pre-target (n_1) and baseline condition (n_2). All trials from the pre-target and baseline conditions were used.

Coh_RIFT.m

How to measure time-resolved tagging response?

1. Select the tagging response sensors
2. Obtain the time-frequency plot of coherence (only for tagging response sensors)
   - Loop over conditions (coherence is calculated over trials)
     - Loop over frequency-of-interest Fi (e.g., [40:1:80]Hz)
       - Obtain ***Hilbert*** analytic signals of the filtered data (Fi ± 5Hz), [MEG + photodiode]
       - calculate coherence between each tagging response sensor and the photodiode, then average coherence over sensors

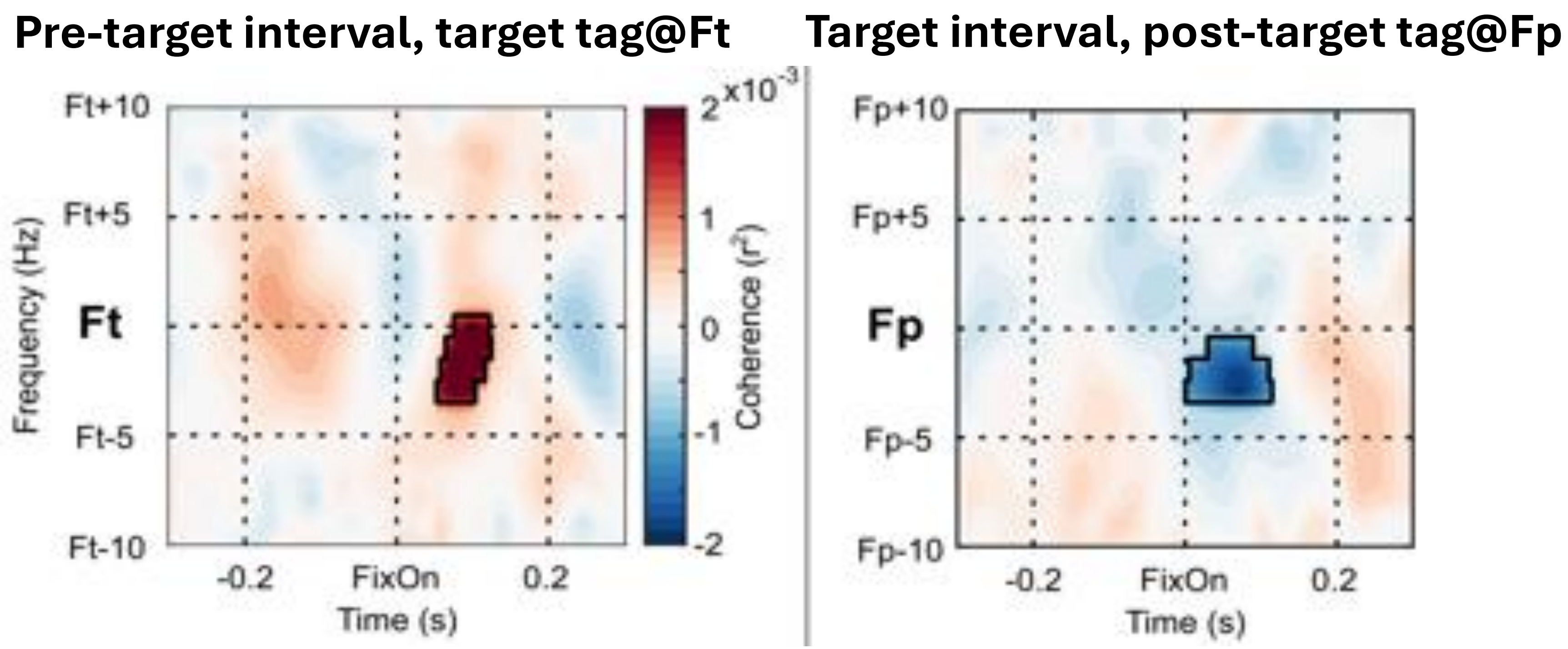
$$coh(t) = \frac{\left|\frac{1}{n}\sum_{j=1}^{n} m_{x_j}(t)\, m_{y_j}(t)\, e^{i\phi_{xy_j}(t)}\right|^2}{\sum_{j=1}^{n} m_{x_j}(t)^2 m_{y_j}(t)^2}$$

where n is the number of trials. For the time point t in the trial j, $m_x(t)$ and $m_y(t)$ are the time-varying magnitude of the analytic signals from a MEG sensor (x) and the photodiode (y) respectively, $\phi_{xy}(t)$ is the phase difference as a function of time (for a detailed description, please see Cohen, 2014).

Group_Coh.m

How to do statistics (on group level)?

Non-parametric permutation test, conditional contrast



(Pan, Frisson, Snell, Federmeier, Jensen, in prep.)

- Doesn’t require a “manual” selection of the time and frequency window, especially convenient when tagging more than one stimulus simultaneously.